# Classifying Root Systems of Semisimple Lie Algebras

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Math 72.02

# Outline


1. Introduction
2. Abstract root systems
3. Dynkin diagrams
4. Consequences
5. Further Reading and References

# Introduction

# Motivation

Say we want to classify Lie groups

- We reduce this problem to classifying simply-connected Lie groups, by first restricting to the identity component, and then passing to the universal cover
- A simply connected lie group is in bijective correspondence with Lie algebras, so we classify Lie algebras
- Lie algebras break down into a "solvable" part and a "semisimple" part. Solvable Lie algebras are hard to classify, but we *can* classify semisimple Lie algebras, and it still gives us a lot of information on the classification problem of Lie groups. Note that semisimple lie algebras are the direct sums of simple lie algebras.

# The adjoint representation

- A lie algebra acts on itself via the adjoint representation $\mathrm{ad}_h(X) = [h, X]$
- In a semisimple Lie algebra $\mathfrak{g}$, let $x : \mathfrak{g} \to \mathfrak{g}$ be the adjoint representation of some element. Then $x = x_s + x_n$
- The Jordan-Chevalley decomposition decomposes a linear operator into a semisimple (diagonalizable) and nilpotent part
- Now, let $\mathfrak{h}$ be a maximal subalgebra consisting only of semisimple elements. This is known as the *Cartan subalgebra*

# Structure theory of semisimple Lie algebras

Here we state, but do not prove, some results about the structure theory of semisimple Lie algebras

- The Cartan subalgebra is self-centralizing. In particular this means it is abelian and its operators commute
- Thus, $\mathfrak{h}$ is *simultaneously diagonalizable*
- This means there is a set of common eigenvectors $X$ such that $\mathrm{ad}_h(X) = \alpha(h)X$ for every $h \in \mathfrak{h}$. The linear functional $\alpha \in \mathfrak{h}^*$ mapping $h$ to its eigenvalue is called a *root*

# The root space decomposition

Given a Cartan subalgebra $\mathfrak{h}$, it holds that $\mathfrak{g}_0 = \mathfrak{h}$ and there is a decomposition (as an $\mathfrak{h}$-module):

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha$$

where $\Phi$ is the set of all nonzero linear functionals $\alpha$ of $\mathfrak{h}$ such that $\mathfrak{g}_\alpha \neq \{0\}$. Moreover, for each $\alpha, \beta \in \Phi$,

- $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subseteq \mathfrak{g}_{\alpha+\beta}$, which is an equality if $\alpha + \beta \neq 0$.
- $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \oplus \mathfrak{g}_{-\alpha} \oplus \mathfrak{g}_\alpha \cong \mathfrak{sl}_2$ as a Lie algebra.
- $\dim \mathfrak{g}_\alpha = 1$; in particular, $\dim \mathfrak{g} = \dim \mathfrak{h} + \#\Phi$.
- $\mathfrak{g}_{2\alpha} = \{0\}$; in other words, $2\alpha \notin \Phi$.
- With respect to the Killing form $B$, $\mathfrak{g}_\alpha, \mathfrak{g}_\beta$ are orthogonal to each other if $\alpha + \beta \neq 0$; the restriction of $B$ to $\mathfrak{h}$ is nondegenerate.

# Root space decomposition - What matters

- Given a complex semisimple Lie algebra, we can find roots $\alpha \in \mathfrak{h}^*$
- We can represent these roots in a *real* subspace of $\mathfrak{h}^*$, and define an inner product on it through the Killing form
- All the other properties give us certain axioms that we will look at in detail in the next section

# Abstract root systems

# Motivation

We can reduce the study of semisimple lie algebras to the study of its root systems

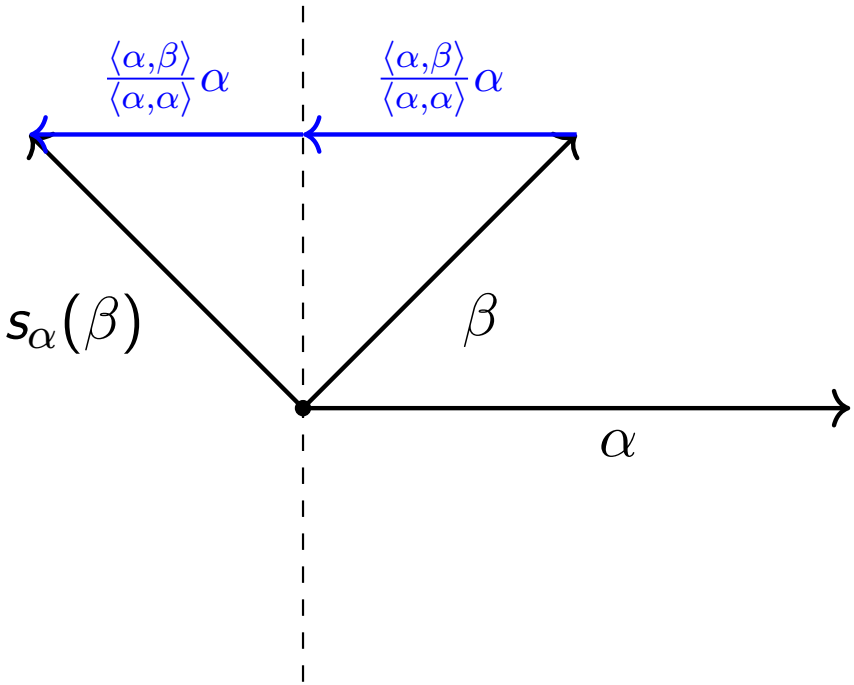
- Each Lie algebra corresponds to a root system, which follows certain rules
- To any set of vectors in Euclidean space that follows those rules, we can assign a Lie algebra
- Thus we can abstract away the fact that these roots come from a Lie algebra, and consider only sets of vectors that satisfy our axioms

## Definition

An *abstract root system* $\Phi$ in a Euclidean space $V$ is a finite set of non-zero vectors (called *roots*) such that:

- $\Phi$ spans $V$.
- For any $\alpha \in \Phi$, the only scalar multiples of $\alpha$ in $\Phi$ are $\pm\alpha$.
- $\Phi$ is closed under reflection: for $\alpha \in \Phi$, the reflection of $\beta$ about the hyperplane perpendicular to $\alpha$, $s_\alpha(\beta) = \beta - 2\frac{\langle\beta,\alpha\rangle}{\langle\alpha,\alpha\rangle}\alpha$, is another root of $\Phi$.
- For $\alpha, \beta \in \Phi$, the quantity $\eta_{\beta,\alpha} = 2\frac{\langle\beta,\alpha\rangle}{\langle\alpha,\alpha\rangle}$ is an integer.

# Illustrating the reflection rules




The reflection rules say that reflecting $\beta$ about the dashed line should give another root. Moreover, it should shift $\beta$ by some integral multiple of $\alpha$

# Angles between roots

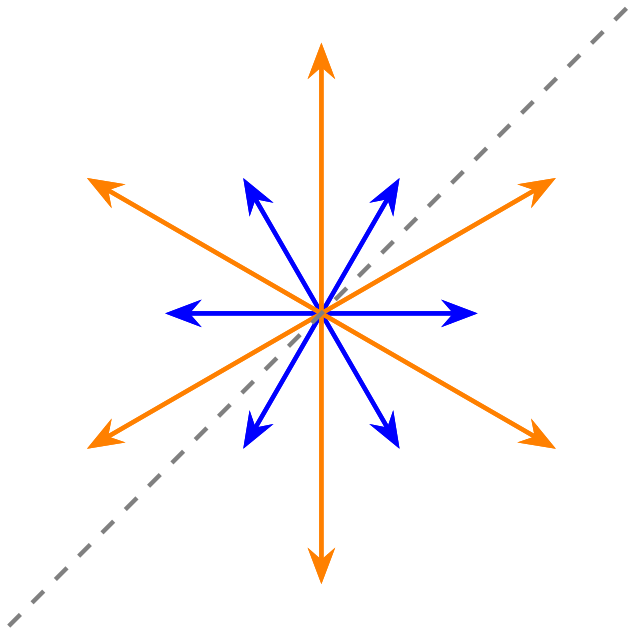
- The reflection rules restrict the angles between roots. Note that $\eta_{\beta,\alpha} = 2\cos(\theta)\frac{||\beta||}{||\alpha||}$ is an integer
- Since $\eta_{\beta,\alpha}$ and $\eta_{\alpha,\beta}$ are both integers, $\eta_{\beta,\alpha}\eta\alpha,\beta = 4\cos^2(\theta)$ is also an integer
- This can only be 0,1, 2, 3, and 4. The case of 4 happens only in the degenerate case $\alpha = \pm\beta$. So the only other possible angles are $30°, 45°, 60°, 90°, 120°, 135°$ and $150°$.

# Reductions

- As we can see, there is a lot of redundancy in the way we have described root systems - we could get rid of some of the vectors and reconstruct the whole root system based on those vectors because of the very rigid rules
- We will now outline a series of reductions
- At the end of this step of reductions, we will get *n simple roots* from which we can reconstruct the root system

# Positive and Negative roots

First, note that every root appears with exactly one negative counterpart. We can thus get rid of half the roots with no loss of information. We arbitrarily call half of them positive, and the other half negative



# Simple roots

**Definition**

Now, call a positive root *simple* if it is not the sum of two other positive roots.

Such a root is bound to exist, because all the positive roots are on one side of a hyperplane, so there must be one "smallest" root. Note that adding two vectors on the same side of a hyperplane gives us a vector further away from that hyperplane on the same side, always.

# Properties of simple roots

Here are the properties of simple roots that we will need when classifying Dynkin diagrams

**Theorem**

*The angle between two simple roots cannot be acute, i.e.,* $\langle \alpha, \beta \rangle \leq 0$

**Corollary**

*The simple roots are linearly independent*

The corollary follows from the theorem by noting that non-acute vectors on the same side of a hyperplane are linearly independent

**Corollary**

*There are exactly n simple roots*

Follows from linear independence

# Proof of the Theorem

## Proof.

Suppose $\langle \alpha, \beta \rangle > 0$, and WLOG suppose $\langle \alpha, \alpha \rangle \geq \langle \beta, \beta$. Now, since $\eta_{\alpha,\beta}\eta_{\beta,\alpha} < 4$, at least one of them is $\pm 1$. Since that one will have the larger denominator, it has to be $\eta_{\beta,\alpha} = 2\frac{\langle \alpha,\beta \rangle}{\langle \alpha,\alpha \rangle}$. Since we assumed $\langle \alpha, \beta \rangle > 0$, this means $\eta_{\beta,\alpha} = 1$

Then, $s_\alpha(\beta) = \beta - \eta_{\beta,\alpha}\alpha = \beta - \alpha$ is a root, and so is $\alpha - \beta$. Exactly one of them is a positive root. If $\beta - \alpha$ is positive, this contradicts the simplicity of $\beta$, and if $\alpha - \beta$ is positive, it contradicts the simplicity of $\alpha$



# Dynkin diagrams

# Dynkin Diagrams

A Dynkin diagram is a graph where:

- Each node represents a simple root
- Edges encode the angle between roots, arrows go from the longer to the shorter root

| Angle between roots | Diagram edge | $\langle \alpha_1, \alpha_2 \rangle$ |
|---|---|---|
| 90° |  | 0 |
| 120° |  | $-\frac{1}{2}$ |
| 135° |  | $-\frac{\sqrt{2}}{2}$ |
| 150° |  | $-\frac{\sqrt{3}}{2}$ |

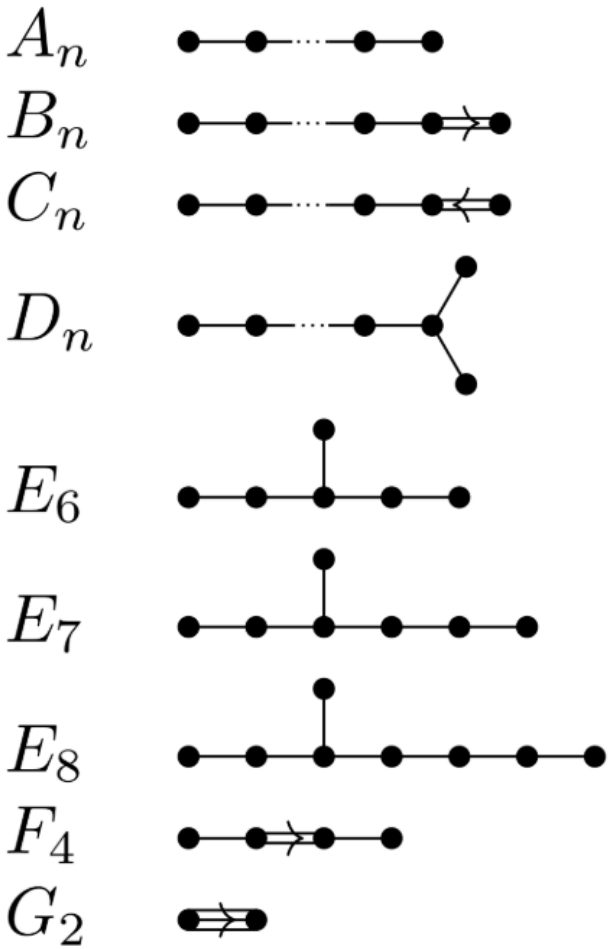
Note that $4\langle \alpha_1, \alpha_2 \rangle^2 = \#$ of lines between $\alpha_1$ and $\alpha_2$

# Admissible Dynkin diagrams

- We want to find all the possible Dynkin diagrams
- Call a Dynkin diagram *admissible* if there exist $n$ vectors in Euclidean space such that their pairwise angles correspond to the angle we get from the diagram
- If two nodes are not directly connected, they represent orthogonal vectors
- We can get away with forgetting about length and arrows, and focusing only on angles. From now on, we shall consider every simple root to be unit-length
- We can also get away with considering only connected graphs, because connected graph components of Dynkin diagrams correspond to irreducible root systems

# The classification

- Our goal is to prove the following result
- Dynkin diagrams (and correspondingly, simple Lie algebras) fall into four infinite families and five exceptions
- These are $A_n$, $B_n$, $C_n$, $D_n$, $E_6$, $E_7$, $E_8$, $F_4$, $G_2$.



# Subdiagram Law

**Theorem**

*Every subdiagram of an admissible Dynkin diagram is again admissible.*

This is true because we can simply forget about some of the simple roots, and the rest of the simple roots still follow the angle constraints

# Tree Law

## Theorem

*An admissible Dynkin diagram contains no cycles. Thus an irreducible Dynkin diagram is a tree.*

*Proof.* Suppose a cycle of simple roots $\alpha_1, \ldots, \alpha_k$ exists. Let

$$v = \alpha_1 + \cdots + \alpha_k.$$

Consider $\langle v, \alpha_i \rangle$ for each $i$. If $\alpha_i$ and $\alpha_j$ are adjacent, $\langle \alpha_i, \alpha_j \rangle < -\frac{1}{2}$. In the cycle, each root is adjacent to at least 2 more, so $\langle v, \alpha_i \rangle \leq 1 - \frac{1}{2} - \frac{1}{2} +$ (negative or 0 terms), so $\langle v, \alpha_i \rangle \leq 0$. Therefore,

$$\langle v, v \rangle = \sum_{i,j} \langle \alpha_i, \alpha_j \rangle \leq 0,$$

but an inner product of something with itself must be positive.

# Valence Law

## Theorem

*Each node has at most 3 lines coming into it*

## Proof

Suppose $\alpha_0$ is connected to $k$ other roots. None of these roots can be connected to each other, as cycles are forbidden. Recall that $\langle \alpha_i, \alpha_j \rangle$ is the number of lines between them. Our claim reduces to showing

$$\sum_{i=1}^{k} 4\langle \alpha_0, \alpha_i \rangle^2 < 4$$

or

$$\sum_{i=1}^{k} \langle \alpha_0, \alpha_i \rangle^2 < 1$$

# Valence Law proof

## Proof (continued)

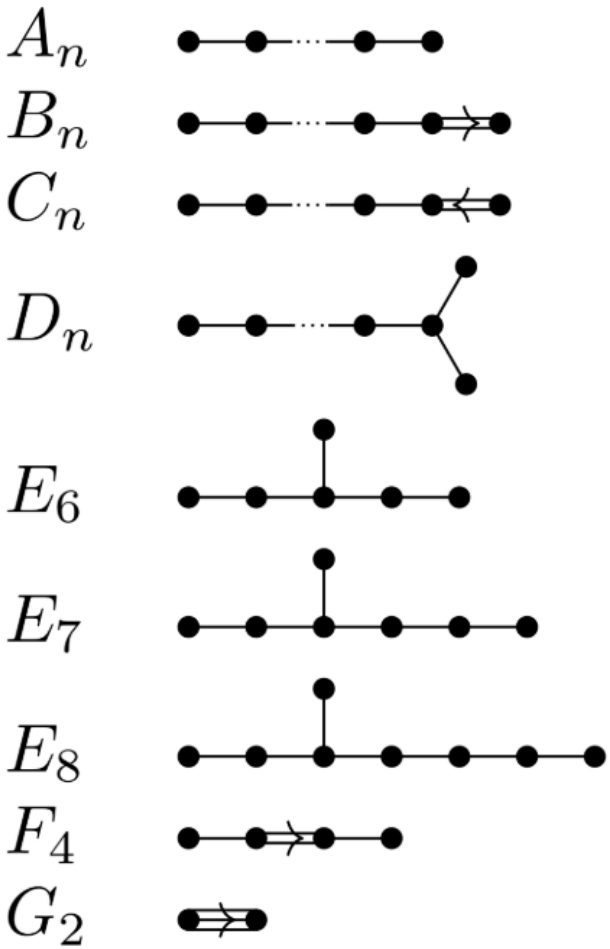
Note that $\langle \alpha_0, \alpha_i \rangle^2$ is the squared length of the projection of $\alpha_0$ along $\alpha_i$, so $\sum_{i=1}^{k} 4\langle \alpha_0, \alpha_i \rangle^2 < 4$ is the squared length of the projection of $\alpha_0$ onto the span of the $\alpha_i$.

Since simple roots are linearly independent, $\alpha_0$ is not in this span and has some component outside. Thus the squared length of the component inside this span is less that the total squared length, 1. Or,

$$\sum_{i=1}^{k} \langle \alpha_0, \alpha_i \rangle^2 < 1$$

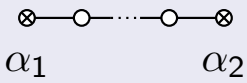
# Reductions

- By the laws we've seen, we can do some reductions
- By the valence law, a triple-bond graph cannot be extended, so the only graph with a triple bond is 



# Contraction law

## Theorem

*Given a string of nodes*  *, we can contract away the simple chain to just*  *, where the contracted root inherits the connectivities at both ends*

## Proof.

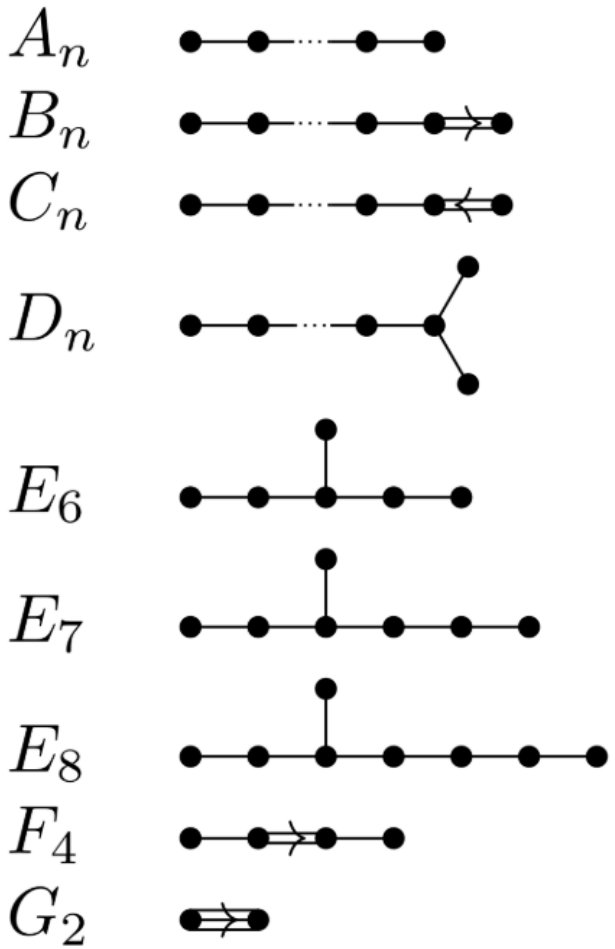
Let the string be $\alpha_1, \ldots, \alpha_n$ where $\alpha_1$ and $\alpha_n$ are the edges. Let $v = \alpha_1 + \cdots + \alpha_n$, then

$$\langle v, v \rangle = n + 2 \sum_{i=1}^{n-1} \langle \alpha_i, \alpha_{i+1} \rangle = n - (n-1) = 1$$

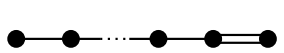
so $v$ is a unit vector. It inherits the connectivities of $\alpha_1$ and $\alpha_n$ because for any $u$, $\langle v, u \rangle$ is either $\langle \alpha_1, u \rangle$ or $\langle \alpha_n, u \rangle$. □

# Reductions

- The contraction law helps a lot. No graph can have two double bonds, or a double bond and a tripod, or two tripods, because we can contract away the path between them and get a valence 4 node
- The only remaining cases are the graphs with a single double bond and the graphs with a single tripod



# Double-bond law

Consider the double bond case. We can explicitly construct solutions to  , ($B_n$ and $C_n$, depending on the arrow direction) so we need to consider cases like  where the double bond is in the middle. We shall use the subdiagram law to show that  is inadmissible, which would show that the only double-bonded diagram with a double bond in the middle is 

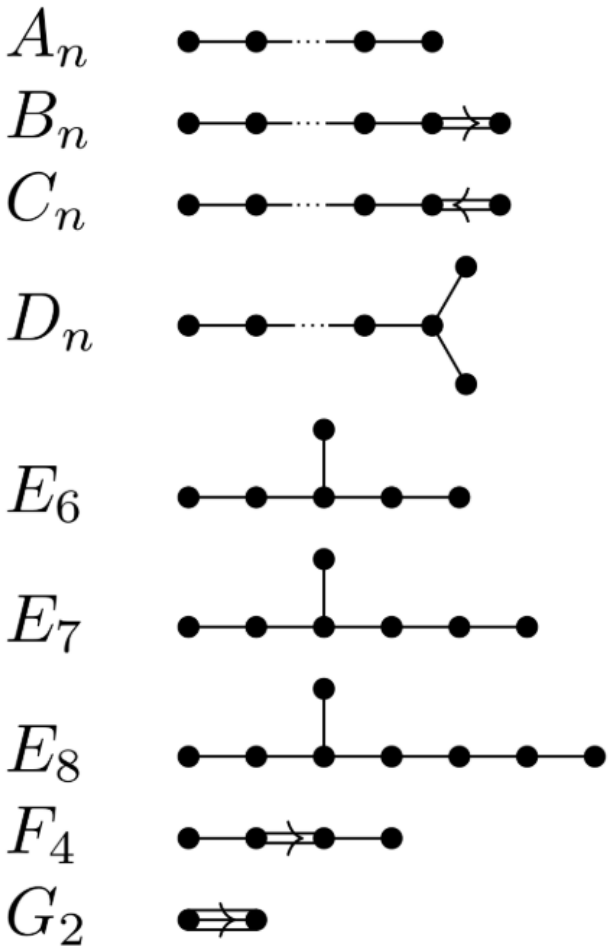
# Proof of the double bond law

## Proof that  is inadmissible

Label the vertices $\alpha_1, \ldots, \alpha_5$, with the double bond between $\alpha_3$ and $\alpha_4$. Now consider $v = \alpha_1 + 2\alpha_2 + 3\alpha_3$ and $w = 2\alpha_4 + \alpha_5$. Then by using the inner products, we can compute that $\langle v, w \rangle^2 = 18$, $||v||^2 = 6$ and $||w||^2 = 3$, which contradicts the Cauchy-Schwartz inequality (which is strict here)

# The classification

- We have characterized all the graphs with a double or triple bond, and showed that they are $B_n, C_n, F_4, G_2$. We need to consider single bond graphs.
- If there are no branches, then we have $A_n$.
- We showed earlier that there can be at most one tripod, so now we consider graphs with a single tripod

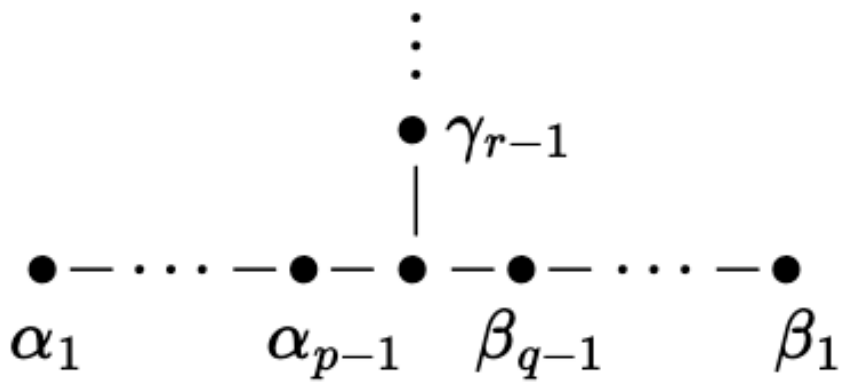


# Tripod Law

## Theorem

*If we have a Dynkin diagram with a single branch point, and the length of the branches (including the center point in each case) are p, q, and r, then*

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$$



# Tripod Law proof outline

**Proof.**

The proof follows the same span-projection idea from the proof of the valence law. Consider three unit vectors $u$, $v$, and $w$ which are linear combinations of vectors from the three branches (without the center point). Then because the central vector, $\eta$, is not in their span, $\langle \eta, u \rangle^2 + \langle \eta, v \rangle^2 + \langle \eta, w \rangle^2 < 1$

If we could construct $u, v$, and $w$ such that $\langle \eta, u \rangle^2 = \frac{1}{2}\left(1 - \frac{1}{p}\right)$ (and analogously for the others), we would get the desired result. We outline this construction for $u$ below. □

# Constructing $u$

Let $u = \frac{u'}{||u'||}$ where

$$u' = \alpha_1 + 2\alpha_2 + \cdots + (p-1)\alpha_p$$

Then

$$\begin{aligned}
||u'||^2 &= \sum_{k=1}^{p-1} k^2 + 2\sum_{k=1}^{p-2} k(k+1)\left(-\frac{1}{2}\right) \\
&= (p-1)^2 + \sum_{k=1}^{p-2} \left(k^2 - k(k+1)\right) \\
&= (p-1)^2 - \sum_{k=1}^{p-2} k \\
&= (p-1)^2 - \frac{(p-1)(p-2)}{2} = \frac{p(p-1)}{2}
\end{aligned}$$

# Finishing the construction

Now, consider

$$\langle \eta, u \rangle^2 = \frac{\langle \eta, u' \rangle^2}{||u'||^2}$$

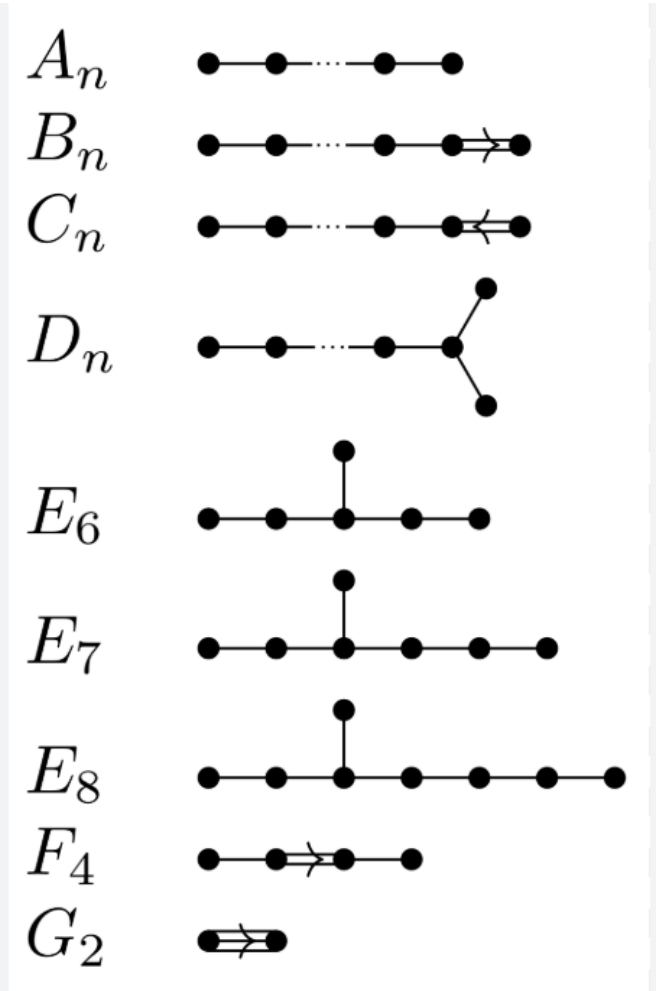
Since $\eta$ is orthogonal to all but $\alpha_{p-1}$, the numerator reduces to $(p-1)^2 \left(-\frac{1}{2}\right)^2$ (where the $-\frac{1}{2}$ comes from the angle), and the denominator is $\frac{p(p-1)}{2}$. Thus

$$\langle \eta, u \rangle^2 = \frac{(p-1)^2 \left(\frac{1}{4}\right)}{\frac{p(p-1)}{2}} = \frac{1}{2}\left(1 - \frac{1}{p}\right)$$

as needed.

# Finishing the classification

- All triple bond graphs follow $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} > 1$
- If any one of the denominators is 1, the equation is true, but then it is not a real triple bond and we get $A_n$
- If two of the denominators are 2, then the equation is true regardless of the third, and we get the infinite family $D_n$
- If all three denominators are greater than 2, then the equation is always false. Thus the best we can hope for is a chain with a single-node "stub." These are the $E_6, E_7, E_8$



# Consequences

# Consequences

- A Dynkin diagram corresponds uniquely to a semisimple complex Lie algebra
- A semisimple complex Lie algebra can correspond to many real forms, but there is a unique "compact form," which gives rise to a compact semisimple Lie group
- Thus every compact, simply-connected, simple Lie group is one of $A_n, B_n, C_n, D_n, G_2, F_4, E_6, E_7, E_8$

# The compact simple simply-connected lie groups

$$
\begin{array}{ccc}
A_n \equiv & SU(n+1) & n \geq 1 \\
B_n \equiv & Spin(2n+1) & n \geq 2 \\
C_n \equiv & Sp(n) & n \geq 3 \\
D_n \equiv & Spin(2n+1) & n \geq 4 \\
G_2 & & \\
F_4 & & \\
E_6, E_7, E_8 & &
\end{array}
$$

# Compact connected Lie groups

We now get a classification of compact connected lie groups

- Passing out of the universal cover is a simple process of modding out by some discrete subgroup of the center
- The non-semisimple portion of a compact Lie group is a torus. From this we get a classification of compact connected Lie groups

Every compact connected Lie group has the form

$$G = \frac{S_1 \times \cdots \times S_m \times T}{Z}$$

where

- $S_i$ are compact simple Lie groups
- $Z$ is a discrete subgroup of the center of the semisimple portion
- $T$ is a torus

# Further Reading and References

# Further reading

- See the accompanying paper for a more thorough description of the reduction from Lie groups to root systems
- Some of the root systems constructed have interesting properties. For instance, the $E_8$ lattice $= \left\{\mathbb{Z}^8 \cup (\mathbb{Z} + \frac{1}{2})^8 \mid \text{the sum of all coordinates is even}\right\}$ shows up in sphere packings and sphere kissing numbers, error correcting codes, modular form theory, and even my research!
- The classification of *real* Lie algebras uses objects called Satake diagrams
- There are many other extensions to Dynkin diagrams, like affine, twisted, etc. that talk about more general objects than semisimple Lie algebras [1]
- Dynkin diagrams can also be used to classify "reflection groups" and "coxeter groups"

[1] Funnily enough, the most comprehensive list of those (other than Wikipedia) is best found in the documentation of the dynkin-diagrams latex package

# References


- *Representation Theory: a First Course*, Fulton and Harris. The whole book builds up to this classification very carefully and rigorously. Esp. see Lectures 9, 14, and 21.
- Terence Tao's blog
- *The Topology of Lie Groups*, Randy S, `https://www.cphysics.org/article/92035.pdf`
- *Seminar on Lie Algebras*, Heidelberg University, `https://matematiflo.github.io/WiSe_2023_2024/LieAlgebrasSeminar.html`